

NETWORK FILE SYSTEMS – [NFS]

❖ What is NFS? [Port NFS - 2049]

- NFS stands for Network File System.
- It's a **distributed file system protocol** that allows users to access files **shared storage using the NFS**.
- It automatically scales based on demand and supports high availability across multiple Availability Zones (AZs).

✓ Key Points:

- **Developed by:** Sun Microsystems in 1984.
- **Used for:** Sharing files and directories between systems over a network.
- **Common in:** Unix/Linux environments, though available on Windows with support tools.
- **Protocol:** Operates over TCP/IP, typically on port **2049**

❖ How NFS Works:

- A **server** exports one or more directories.
 - A **client** mounts those directories over the network.
 - The mounted directories appear as part of the client's file system.
- **The protocol handles:**
- File access (read/write)
 - Directory traversal
 - File locking (with appropriate version support)
 - User authentication (in advanced versions)

Communication is typically done using Remote Procedure Calls (RPCs)

✓ Typical Use Cases

1. **Centralized File Storage** :- Multiple clients need access to a **shared file system**.
2. **Home Directory Sharing** :- Users' home directories are stored on a **central server**.
3. **Web Server Clustering** :- Multiple web servers share the **same content** (e.g., HTML, images, videos).
4. **Application Data Sharing** :- Applications (e.g., databases, analysis tools) share logs or datasets over NFS.
5. **Backup and Archiving** :- Backup servers mount file systems via NFS to copy data for backup.
6. **Virtual Machine Storage** :- Hypervisors or containers use NFS for persistent storage.
7. **Development Environments** :- Developers access and modify shared codebases or environments.

❖ What is Amazon EFS?

- **Amazon EFS** stands for **Amazon Elastic File System**. It's a **fully managed, scalable, and elastic network file system** offered by **AWS (Amazon Web Services)**, designed to be used with Amazon EC2 instances and other AWS services

✓ Key Features of Amazon EFS:

- **Fully managed** ➤ AWS handles provisioning, patching, scaling, and availability
- **Elastic** ➤ Automatically scales up or down as files are added or removed
- **Shared access** ➤ Multiple EC2 instances or containers can access the file system at the same time
- **Standard NFS support** ➤ Compatible with NFSv4 and NFSv4.1 protocols
- **Durable and highly available** ➤ Data is stored across multiple Availability Zones (AZs)
- **Secure** :- Support IAM, VPC security groups, encryption at rest/in transit

❖ **Real-Time Use Cases of Amazon EFS**

- **Use Case** ➤ Real-Time Advantage
- **Web hosting** ➤ Shared, live-updated content
- **Microservices (ECS/EKS)** ➤ Shared config/logs at runtime
- **Media processing** ➤ Concurrent read/write for large files
- **ML/Analytics pipelines** ➤ Centralized, simultaneous data access
- **Developer environments** ➤ Shared tools and codebases
- **Home directories** ➤ Persistent user sessions
- **CI/CD pipelines** ➤ Shared build/workspace folders

❖ **Comparison Table: EFS vs EBS**

Feature	Amazon EFS	Amazon EBS
Type	Network File System (shared file storage)	Block Storage (attached to a single instance)
Access	Multiple instances (shared via NFS)	One instance at a time (except multi-attach)
Protocol	NFSv4.0 / NFSv4.1	Raw block-level I/O
use Case	Shared storage for Linux-based workloads	Boot volumes, databases, low-latency apps
Scalability	Auto-scales up/down with usage	Fixed size; must resize manually
Durability & Availability	Multi-AZ, highly available	AZ-specific unless backed up
Performance Modes	General Purpose, Max I/O	Varies by volume type (gp3, io1, etc.)
Cost Model	Pay per GB used	Pay per provisioned GB
Mounting Requirement	Mount via NFS client	Attach as a disk to EC2

❖ **How does EFS handle scaling?**

1. EFS **automatically scales** up or down based on the amount of data stored.
2. **No need** for manual provisioning or resizing.
3. It provides instant access to additional storage as needed

❖ **What security measures can be implemented in Amazon EFS?**

1. Use **IAM** (Identity and Access Management) **policies** to control access.

2. Attach **security groups** to allow traffic only from specific **EC2 instances**.
3. Enable encryption at rest using AWS Key Management Service (KMS).
4. Enable encryption in transit using **TLS** (Transport Layer Security).
5. Use AWS **VPC** and **NACLs** (Network ACLs) to restrict access.

❖ How can an EC2 instance mount an EFS file system?

1. Install the necessary NFS client:

`sudo yum install -y amazon-efs-utils` # For Amazon Linux 2 / RHEL
`sudo apt install -y nfs-common` # For Ubuntu/Debian

2. Create a mount point:

`sudo mkdir -p /mnt/efs`

3. Mount the EFS file system:

`sudo mount -t efs fs-xxxxxxx:/ /mnt/efs`

4. To make it persistent, add an entry to `/etc/fstab`:

`fs-xxxxxxx:/ /mnt/efs efs defaults,_netdev 0 0`

❖ What are the throughput modes in Amazon EFS?

■ Bursting Throughput Mode(Default):

1. Throughput **automatically scales** with the size of your file system.
2. Through put rate You get **1 MiB/s per GiB** of data stored, up to a limit.

■ Provisioned Throughput Mode:-

1. You manually set the throughput **independent** of the file system size.
2. Choose a **fixed throughput rate (in MiB/s)**, even if your data storage is small.

❖ How Is Amazon EFS Billed?

- Amazon EFS (Elastic File System) is billed based on **usage** and selected **performance (measured in GB/month)**.
- **Optional costs** for provisioned throughput and backup storage

❖ What monitoring tools can be used for Amazon EFS?

- **Amazon CloudWatch**:- Provides metrics, logs, and alarms for EFS performance and usage.
- **AWS CloudTrail**:- Logs API calls related to EFS (e.g., create, delete, modify file systems).
- **Amazon Trusted Advisor**:- Provides optimization and security checks for EFS usage.

[General Interview Questions]

Q1. What is Amazon EFS?

Q2. How does Amazon EFS differ from Amazon EBS?

Q3. What are the key use cases of Amazon EFS?

Q4. What are the main performance modes available in Amazon EFS

➤ General Purpose (GP) Performance Mode

- **Default mode** when you create a file system.
- optimized for **low-latency-sensitive** use cases.
- **Suitable for**:
 - Web serving
 - Content management systems (CMS)
 - Development/test environments

- **Max I/O Performance Mode**
- Best for high **throughput workloads**.
- *Suitable for:*
 - Big data analysis
 - Media processing
 - Machine learning workloads

Q5. How does EFS handle scaling?

- **EFS automatically** scales up or down based on the amount of data stored.
- No need for manual provisioning or resizing.
- It **provides instant access to additional** storage as needed.

Q6. What security measures can be implemented in Amazon EFS?

- Use **IAM (Identity and Access Management)** policies to control access.
- Attach **security groups** to allow traffic only from specific EC2 instances.
- Enable **encryption at rest** using AWS Key Management Service (KMS).
- Enable **encryption in transit** using **TLS (Transport Layer Security)**.
- Use **AWS VPC and NACLs (Network ACLs)** to restrict access.

Q7. What are the throughput modes in Amazon EFS?

- **Bursting Throughput Mode(Default):** Uses a credit-based system to allow high burst speeds when needed.
- **Provisioned Throughput Mode:**
 - Allows users to set a fixed throughput, independent of the storage size.
 - Best for applications with **consistent high throughput needs**.

Q8. How is Amazon EFS billed?

- Charged **based on actual storage used** (measured in GB/month).
- No upfront costs or provisioning fees
- Optional costs for provisioned **throughput** and **backup storage**.
- **Intelligent-Tiering** moves infrequently accessed files to a lower-cost storage class.

Q9. What monitoring tools can be used for Amazon EFS?

- **Amazon CloudWatch:** Monitors storage usage, I/O requests, and throughput. AWS
- **CloudTrail:** Tracks API activity related to EFS.
- **Amazon Trusted Advisor:** Provides recommendations for optimizing EFS costs